

[Indeksi](#)[Intelligent Field Devices \(IFDM\)](#)[Labra 1](#)

 [Homework: Control Valve Videos](#) ▾
Due: Sunday, 4 October 2026, 5:42 PM

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 [Easter Egg: Pipe Run](#) ▾



 [Valve Structures, Actuators, Positioners](#) ▾

 [Valmet Neles ND9000 Positioner Manual](#)

 [Pump Diagram Sulzer \(140 Impeller\)](#) ▾

 [Pump Diagram Kolmeks \(105 Impeller\)](#) ▾

 [Know Pulp \(Pumps\)](#)

 [Nelprof Sizing Tool](#)

 [PI-kaavio](#) ▾

 [Valve Sizing](#)
Due: Monday, 19 October 2026, 3:00 PM

To do ▾

CONTROL VALVE SIZING ASSIGNMENT

Field Devices and Instrumentation

Objective

The objective of this assignment is to learn the fundamentals of control valve sizing by performing a complete sizing exercise independently. During the assignment, students will become familiar with the valve sizing process, the use of sizing software, and the analysis and interpretation of sizing results. The knowledge and skills acquired during the assignment are also essential for the course examination.

Assignment Requirements

1. The assignment is completed **individually**. Each student sizes a control valve for their assigned process application. The process data and operating conditions are provided in the accompanying Excel file.
2. The final outcome is a control valve selected and sized for the process application. The valve selection must be justified using calculations, sizing software results, graphical outputs, and supporting technical documentation.

Report Preparation

The report shall be prepared using the JAMK reporting template and following the JAMK APA 7 referencing guidelines. [\[Sa Lhteid...tointiohje | Other Sites\]](#), [\[Other repo...plate.dotx | Word\]](#)

Please note the following:

1. Use the JAMK reporting template and preserve the original formatting. [\[Other repo...plate.dotx | Word\]](#)
2. An abstract page is **not required**.
3. A **table of contents** is recommended, even for a short report.
4. **References** shall be presented according to the JAMK APA 7 guidelines. [\[Sa Lhteid...tointiohje | Other Sites\]](#)
5. All figures and graphs must be referenced and discussed in the text.

Assignment Procedure

1. Familiarise Yourself with the Process Application

- Study your assigned process application.
- Identify and collect the required sizing data.

2. Review the Pump Curve

- Analyse the pump performance curve.
- Estimate the maximum available pressure.

3. Calculate Pressure Loss Between the Pump and the Valve

- Identify and document all components that cause pressure loss between the pump and the valve.
- Estimate the total pressure loss in the system section under consideration. You may use an online pressure [drop calculation tool](#) if appropriate.

4. Calculate the Pressure Drop Across the Valve at the Design Operating Point

- Use the D_{Pm} formula described in the *Flow Control Manual*.

5. Perform Valve Sizing

- Enter the process data into the sizing software.
- Perform the initial sizing using an **RE-series segmented ball valve**.

6. Analyse the Sizing Results

- Study the sizing software graphs and results.
- Compare the results with the examples presented in the *Flow Control Manual*.

7. Perform a Comparative Evaluation

- Repeat the sizing exercise using an **M1-series ball valve**.

8. Select the Most Suitable Valve

- Decide whether the RE-series or M1-series valve is more suitable for the application.
- Justify your selection using sizing results, graphs, and technical analysis.

9. Review Product Documentation

- Study the manufacturer's technical documentation for the selected valve.
- Use relevant information from the documentation to support your report.

10. Document All Work

Include the following in your report:

- Process description
- Input data
- Calculations
- Assumptions
- Sizing software outputs
- Graphs and figures (from Nelprof)
- Analysis of results
- Valve selection rationale
- Conclusions

Sources

At a minimum, use the following sources:

- *Flow Control Manual*
- Valve manufacturer documentation (e.g., Valmet)
- Sizing software reports and graphical outputs

All sources shall be cited according to the JAMK APA 7 guidelines. [\[Sa Lhteid...tointiohje | Other Sites\]](#)

Assessment

The assignment is assessed on a **Pass / Fail** basis.

To receive a passing grade for the assignment, the student must demonstrate the ability to:

- collect and analyse the required process data,
- perform and document valve sizing calculations,
- use valve sizing software appropriately,
- interpret sizing graphs and results,
- compare alternative valve solutions, and
- justify the final valve selection using technical reasoning.

Completion of the assignment is a mandatory course requirement. The assignment does not directly affect the numerical course grade, but it must be completed successfully before the examination results can be used in the final course assessment.

Assessment Criteria

Assessment Area	Pass	Fail
Report structure and formatting	Report follows JAMK reporting guidelines and the required template is used correctly.	Report structure is incomplete or formatting requirements are not followed.
Process application description	Process application and sizing data are presented clearly and supported with appropriate figures.	Process description or sizing data are missing, unclear, or incorrect.
Calculations and sizing methodology	Pressure losses and D _{Pm} calculations are presented and all relevant components are considered.	Calculations are missing, incorrect, or important components are omitted.
Use of sizing software	Both valve alternatives have been sized and the relevant graphs are included and referenced.	Sizing software has not been used appropriately or required results are missing.
Analysis of results	Results are analysed critically and valve alternatives are compared using technical reasoning.	Analysis is missing, inadequate, or technically incorrect.
Use of references	Relevant sources are used and cited according to APA 7 guidelines.	References are missing, incorrect, or inadequately cited.

Conclusions and valve selection	The final valve selection is justified clearly and logically based on calculations and analysis.	Conclusions are missing or the valve selection is not justified.
Use of AI tools	Any use of AI tools supports the student's own learning and analysis. The report demonstrates understanding of the process, calculations, sizing results, and conclusions. The student is able to justify the presented work.	The report appears to be predominantly AI-generated, contains technical inaccuracies or hallucinated content, or does not demonstrate the student's own understanding of the valve sizing process and results.
Note: Failure to address any major assessment area may result in a failing grade. Minor deficiencies do not automatically lead to failure if the overall work demonstrates adequate competence. The assignment is intended to develop practical skills in control valve sizing and to support preparation for the course examination.		
Use of Artificial Intelligence Generative AI tools may be used to support writing, language checking, and information retrieval. Any use of AI must be documented in the report , including a brief description of how the tool was used and for which tasks. The student remains fully responsible for the technical accuracy of all calculations, analysis, results, and conclusions. Control valve sizing is a technical engineering task in which AI tools may generate incorrect, misleading, or hallucinated information. The submitted work must clearly demonstrate the student's own understanding of the valve sizing process and the interpretation of the results. If the report appears to be predominantly AI-generated, contains unverified AI-generated content, or fails to demonstrate the student's own learning and engineering judgement, the assignment may be rejected. The use of AI must support, not replace, the student's own analysis and learning.		

Submission

1. Submit the report to Moodle **before the exam**.

2. Completing the assignment will significantly support your preparation for the examination.
- Sources: Sa Lähteiden käyttö (APA7-ohje, syksystä 2024 opintonsa aloittaneet) - Opinnäytetyön raportointiohje, [Other reporting template.dotx \(Sa Lähteid...ointiohje | Other Sites\)](#) [\(Other repo...plate.dotx | Word\)](#)

 [Valve positions for students](#) 

 [P&I Diagrams](#)

Additional Material

- [Lessons in Industrial Instrumentation : Tony R. Kuphaldt : Free Download, Borrow, and Streaming : Internet Archive](#)
- [Virtaustekniikka : vedensiltojärjestelmien toiminnallinen suunnittelu nykyaikaisin menetelmin | Jyväskylän ammattikorkeakoulu | Janet Finna](#)
- [Flow technology : modern functional design of water transmission systems | Jyväskylän ammattikorkeakoulu | Janet Finna](#)
- [Säätöventtiilidiagnostiikan analysointi ja visuaalisointi pilvipalvelussa : Control valve PARAMeter ANALyzer 2.0 - Theseus](#)

-  [Laboratory 3d model](#)

-  [Control Valve Virtual Lab](#) 

Control Valve Virtual Lab: getting started

Control Valve Virtual Lab is a browser-based simulator of a water process: a pump feeds water through a quarter-turn control valve into a tank. You will learn to size and select a control valve, see how the valve and the process interact, and see how the same loop behaves under flow control, level control and cascade control. The lab follows chapters 1–3 of the Neles/Valmet Flow Control Manual. Nothing needs installing, and it works in any modern browser.

1. Getting started

- Open the lab link. The page opens in Finnish. Use the **FIN / ENG** buttons at the top right to switch the language. Tooltips and all texts change with it.
- The **Light / Dark** button next to them switches the theme.
- Press **Run / Pause** to start or stop the simulation and **Reset** to return to the initial state. The speed selector makes the simulation run faster, which is handy for slow tank level changes.
- Hover over any parameter name, indicator or tag to read a short explanation (tooltip).

2. The parts of the view

- **Course steps** (the pills at the top): ready-made scenarios, such as Base case, Oversized valve, Undersized valve / weak pump, Valve types, Pressure ratio DPm, Mechanical error, Level control, Cavitation and choked flow, and Cascade control. Selecting one loads suitable parameters and shows a short description and what to observe. Always read the description first.
- **P&ID diagram**: pump P-101, control valve FV-101, flow transmitter FIC-101, tank with level controller LIC-102 and drain valve HV-103. Click a tag to open its **faceplate**.
- **Faceplates** (DCS-style control windows): you can drag them around the screen. Each has a **MAN / AUTO** button (and **CAS** for the level controller where applicable) and an **SP / OP** entry:
 - **MAN**: you set the valve opening (OP) directly.
 - **AUTO**: the controller holds the setpoint (SP) by itself.
 - **CAS** (cascade): the level controller LIC-102 sets the setpoint of the flow controller FIC-101, which in turn moves the valve.
- **Indicator strip**: the key figures at a glance (flow, Δp over the valve, DPm, installed gain, pressure and velocity checks). Values turn to warning colours when a recommended limit is exceeded.
- **Trends**: time plots of flow, level, valve opening and setpoints. Use these to see how the process responds to changes.
- **Analysis tabs**:
 - **Characteristics**: inherent and installed characteristics, with the ideal reference curves.
 - **Sizing**: Kv/Cv calculation and the valve opening at the operating points.
 - **Diagnostics**: checks against the manual's recommendations, with a short verdict.
- **Parameters** (side panel), grouped as Controller, Valve, Operating conditions (sizing), Actuator and positioner, Process, Measurement and Test wave. Test wave lets you add a step or wave to the setpoint so you can study the response in a controlled way.

3. Key quantities

- **Kv / Cv**: the valve's flow coefficient (the larger it is, the more flow at the same pressure drop).
- **Δp**: the pressure drop across the valve. **DPm** is the share of the pressure drop that the valve takes of the total available in the line. A low DPm means that the valve loses its ability to control.
- **G (installed gain)**: how much the flow changes per change in valve opening. Aim for **Gmax / Gmin < 2** and **G ≥ 0.5** across the range, so the loop behaves similarly at all operating points.
- **FL, z, R**: the valve's pressure recovery factor, the cavitation limit coefficient and the rangeability. They are fixed for each valve type and are shown read-only under the valve type.
- **xF, ΔpT**: cavitation indicators. Once ΔpT is exceeded the valve cavitates, and with choked flow the flow no longer increases when you increase the pressure drop.
- **Inlet velocity**: checked against the recommended limits (10 m/s, 7 m/s for butterfly valves) at the selected DN.

4. First experiments

1. **Base case**: start the simulation. In MAN, change OP and see how the flow follows. Then switch FIC-101 to AUTO and change SP.
2. **Oversized valve**: select the step and compare it with the base case. Watch DPm and the installed characteristic. Why does the valve only work in the first 10–20 % of its stroke?
3. **Valve types**: compare segment ball, ball, butterfly, eccentric rotary plug and others. How does the choice of valve change the installed characteristic?
4. **Mechanical error**: increase the dead band or friction and watch the limit cycle on the trend. What does this do to the quality of control?
5. **Level control**: compare direct level control (LIC AUTO) with flow control. Introduce a disturbance, for example by changing the drain valve HV-103 or the pump speed.
6. **Cavitation and choked flow**: reduce the downstream pressure and watch xF, ΔpT and the flow.
7. **Cascade control**: select the step, set the loop to CAS and reduce the pump speed to about 88 %. Then do the same in LIC AUTO. Compare how much the level dips: the cascade keeps the level much steadier, because the inner flow loop corrects the disturbance before it reaches the tank.

5. Remember

- This is a **simplified model**, intended for learning, and it covers water only.
- The FL and z values are **typical** values for each valve type. For real design, use the manufacturer's data sheets and sizing software.
- Experiment freely: change one parameter at a time, observe the trends and indicators, and explain what you see using the manual's equations.

Good luck with the lab!